

EFFECTS OF MESSAGE FRAMING AND EXEMPLARS ON PROMOTING ORGAN DONATION^{1, 2, 3}

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Summary.—People in many countries are unwilling to donate organs. Drawing on previous research regarding the use of message framing and the theory of exemplification promoting intentions to donate organs, this study examined messaging strategies. This study used a 2×2 between-subjects factorial design to examine the joint effects of gain/loss frames and statistical/exemplar appeals on the intentions of 189 Taiwanese college students (108 women, 81 men; age range = 19–24 yr., $M = 21.6$, $SD = 2.9$) regarding organ donation. Each participant was randomly assigned to read one of four versions of an organ donation promotional message and then to complete a questionnaire. The analysis of variance showed a significant interaction between the two factors. Loss-exemplar messages elicited significantly more positive intentions toward donation than did loss-statistical messages. There was no significant difference between the statistical and exemplar appeals observed under the gain-framed condition. The practical implications of developing effective organ donation promotional materials and the limitations of this study are discussed.

In almost every country, the supply of organs for transplantation does not meet the increasing demand (Rudge, Matesanz, Delmonico, & Chapman, 2012; van Dalen & Henkens, 2014), and the organ donation rate in Taiwan is particularly low. The average organ donation rate in Taiwan is about seven donors per one million individuals (Taiwan Organ Registry and Sharing Center, 2015), which is much lower than in European countries or the United States (about 20 per one million individuals; International Registry in Organ Donation and Transplantation, 2015). In this context, it has been imperative to find effective strategies to promote organ donation. Studies have indicated that widespread dissemination of organ donation promotional materials are positively correlated with the number of organ donors, but this correlation was not significant (Stefanone, Anker, Evans, & Feeley, 2012). The graphics and text of materials promot-

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³The English in this document has been checked by at least two professional editors, both native speakers of English. For a certificate, please see: <http://www.textcheck.com/certificate/9MLKkB>.

ing organ donation should be sufficiently persuasive so that those exposed to them will be significantly more willing to donate organs. A study of 53 posters from 15 countries found that although most of the graphics and text of these materials were directly and clearly presented, they did not offer promotional and persuasive organ donation messages to their audience (Chien, 2014). Consequently, it is important to develop effective and persuasive strategies to improve and expand donor recruitment. Thus, this study sought to explore persuasive promotional messages regarding organ donation based on established concepts of message framing drawn from the field of communication and the theory of exemplification.

Message Framing

Message framing is based on prospect theory (Kahneman & Tversky, 1979), an important theoretical construct related to persuasion that refers to the fact that people can be persuaded to make particular decisions through different approaches to an issue that use the same logic to promote a position. When a message is positive and emphasizes the benefits that could be accrued or the disadvantages that could be prevented by taking certain actions, it is said to be "gain framed." When a message is negative and the emphasis is on the disadvantages that could result from certain actions or the losses that could ensue by not taking other actions, it is a "loss-framed" message.

Rothman, Bartels, Wlaschin, and Salovey (2006) proposed a message-framing model for health-related communication, suggesting that people tend to avoid risk when exposed to positive descriptions, whereas they tend to engage in risk when exposed to negative descriptions. For example, gain-framed messages can be used to encourage people to undertake minimally risky protective measures (e.g., using sunscreen) by underscoring the prevention of skin damage (Detweiler, Bedell, Salovey, Pronin, & Rothman, 1999). On the other hand, loss-framed messages can be used to encourage people to undergo risky detection measures; for example, mammography carries the risk that lumps will be detected, but it also comes with the assurance that no negative health consequence will occur because of the examination (Goodall & Appiah, 2008).

Exemplification

Exemplification theory, which is based on cognitive heuristics (Gigerenzer, 1991), suggests that qualitative evidence focused on particular exemplar characters be presented in the context of narratives, such as personal anecdotes, analogies, examples, stories, and testimony. Such qualitative information regarding a given health behavior will be more persuasive than will quantitative evidence regarding that same behavior (Brosius & Bathelt, 1994).

Explanations asserting the persuasiveness of exemplars reference their ability to increase vividness (e.g., Nisbett & Ross, 1980) and their inclusion of an availability heuristic (Kahneman & Tversky, 1973). Zillmann (2006) suggested that the presentation of exemplar messages allows people to group themselves with others of their kind (e.g., a binge drinker as an example of those who consume alcohol). Assessments about health risks and protective behaviors may be changed by identifying with an exemplar of a particular group. AQ:1

Additionally, one study suggested that people might be less critical of exemplar (Kazoleas, 1993), and another study mentioned that an exemplar message may signal that the information is less hostile or threatening because the persuasive intent is masked and the perceived risks are subtle (Moyer-Gusé & Nabi, 2010). Any situation in which a specific example is presented may render a particular piece of media content more memorable or influential (Kopfman, Smith, Yun, & Hodges, 1998). AQ:2

*Gain/Loss Frames and Statistical/Exemplar Appeals
Regarding Organ Donation*

A few studies have examined the differing effects of gain- and loss-framed arguments on intentions toward organ donation. Reinhart, Marshall, Feeley, and Tutzauer (2007) found that students reading a gain-framed message reported significantly more favorable reactions toward the scripted message and lower levels of psychological resistance than did students reading a loss-framed message. Purewal and van den Akker (2010) found that gain-framed messages significantly increased the intentions of British women to donate oocytes compared with loss-framed messages, but no significant difference was observed with regard to the effect of message framing on oocyte donation intentions among Southeast Asian women. Cohen (2010) reported that the gain/loss frame was not a significant contributor to university students' intentions regarding organ donation. The study conducted by Reinhart, *et al.* (2007) reported an effect size of $r = .20$, and that conducted by Purewal and van den Akker (2010) reported an effect size of $r = .12$. Thus, the gain-framed messages in these two studies had small effect sizes.

Several studies have experimentally examined the effects of statistical and exemplar messages on participants' intentions toward donating organs. Studts, Ruberg, McGuffin, and Roetzer (2010) studied the relationship between marrow donation and statistical versus exemplar messages and found that exemplar messages significantly increased medical students' intentions to register with the National Marrow Donor Program compared with statistical messages. Similarly, an organ donation study found that exemplar messages designed to increase participants' willing-

ness to donate organs were evaluated more positively than were statistical messages (Feeley, Marshall, & Reinhart, 2006). Weber and Martin (2006) examined the effects of message content (exemplar, statistical, and none), and the results of a chi-squared analysis indicated that exemplar messages were more effective than were statistical and no messages for promoting organ donation. The studies conducted by Studts, *et al.* (2010), Feeley, *et al.* (2006), and Weber and Martin (2006) reported small effect sizes of $r = .16$, $r = .20$, and $r = .18$, respectively.

Several studies have argued that combining a message-framing approach with exemplar appeals may offer a more relevant and easily processed message context (Zillmann, 1999; Zillmann & Brosius, 2000) because the elaboration of the negative or positive effects of health-related behaviors may be enhanced through the affective reactions generated by a narrative (Zillmann, 2006). Research examining the combined effects of message framing and exemplar messages on organ donation is scarce. One study (Sun, *in press*) examined the combined effects of message framing and rational/emotional appeals on Korean participants' intentions toward organ donation. The results showed a significant interaction effect on the intention to donate an organ. Participants who read gain-framed messages with rational appeals showed a more positive attitude than those who read gain-framed messages with emotional appeals. In contrast, participants who read loss-framed messages with emotional appeals showed a more positive attitude than those who read loss-framed message with rational appeals.

Limited research regarding whether gain- or loss-framed messages combined with statistical/exemplar messages are more effective in communicating organ donation information has been conducted. Consistent with Sun (*in press*), the following hypothesis was tested in this study:

Hypothesis 1. A participant's intention to donate an organ will differ depending on the type of message framing and the use of statistical or exemplar appeals.

METHOD

Participants

The sample consisted of 189 college students recruited from undergraduate classes at a large public university in Taiwan. The participants received extra credit in exchange for taking part. The sample included 108 women and 81 men ages 19–24 yr. ($M = 21.6$, $SD = 2.9$). None of the participants had signed an organ donor card or registered as an organ donor.

Measure

Two items assessed the strength of the intention to register as a donor and to discuss the decision with relevant individuals: "It is likely that I

will register my consent to donate my organs upon death" (discuss my donation decision with my partner or family members) and "I intend to register my consent to donate my organs upon death" (discuss my donation decision with my partner or family members). Both items were rated on a scale with anchors 1: Strongly disagree and 7: Strongly agree (Hyde & White, 2009), and ratings on the two items, which were significantly correlated ($r = .79$), were summed to create an intention scale.

Procedure

The participants were invited to take part in this study while they were attending a regular class. They then completed a questionnaire on demographic characteristics. After completing the questionnaire, the participants were randomly assigned to read one of four versions of an organ donation promotional message. They then answered a questionnaire concerning their intentions regarding organ donation. The final portion of the questionnaire consisted of the manipulation check.

Design

The current experiment used a 2×2 between-subjects factorial design, with the gain/loss frame as one factor and the statistical/exemplar appeal as the second factor. The participants read one of four messages about organ donation. The four different combinations of messages (Table 1) examined in this study included: (1) a gain–statistical message, (2) a loss–statistical message, (3) a gain–exemplar message, and (4) a loss–exemplar message. Statistical messages emphasized numbers, and exemplar messages vividly depicted an individual's story. Gain-framed messages conveyed the idea that patients would survive due to organ transplants, whereas loss-framed messages conveyed the idea that patients would die if donated organs were not available. The stimulus materials were successfully pretested using a within-subjects design with a smaller group of participants to ensure that the desired manipulations were clearly perceived by message recipients.

Analysis

An analysis of variance (ANOVA) was performed to address Hypothesis 1. After the controlled experiment, two questions were used to test the effectiveness of each of the four statements that conveyed the manipulations. Independent-sample t tests were used to ensure that the manipulation of the features of the message was successful. Participants who received the gain-framed message were significantly more likely to view the tone of the statement as positive ($M = 4.32$, $SD = 1.48$) than were those exposed to the loss-framed message ($M = 2.57$, $SD = 1.32$; $t_{187} = 8.60$, $p < .01$). Participants who received the statistical messages ($M = 4.22$, $SD = 1.32$) were significantly more likely to agree with the statement "The message relied

TABLE 1
THE FOUR VERSIONS OF PERSUASIVE MESSAGE USED IN THIS STUDY

	Gain-Statistical Message	Gain-Exemplar Message
Slogan message content	The gift of love can prolong life, Currently, about 8,000 people are waiting for organ transplants. About 200 people donate their organs annually. There are still more than 7,000 people waiting for miracles , but their chances of receiving one are extremely low. If more people donated, more people would be reborn because of the availability of organ transplants.	The gift of love can prolong life. Mrs. Tsai's daughter died in a car accident, and Mrs. Tsai donated her daughter's organs. Hui was lucky to receive a heart from Mrs. Tsai's daughter. Hui and Mrs. Tsai therefore became very good friends, and Mrs. Tsai seems to have a daughter again. If more people donate, more people like Hui will be reborn because of the availability of organ transplants.
	Loss-Statistical Message	Loss-Exemplar Message
Slogan message content	Life will wither without the gift of love. Currently, about 8,000 people are waiting for organ transplants. About 200 people donate their organs annually. There are still more than 7,000 people who are seriously ill , and the proportion of them who receive transplants is still extremely low. If more people do not donate, many people who are waiting for an organ transplant will die .	Life will wither without the gift of love. Mrs. Tsai's daughter died in a car accident, and Mrs. Tsai thought about donating her daughter's organs. However, she decided not to donate her daughter's organs. Unfortunately , Hui did not receive a heart from Mrs. Tsai's daughter and passed away. If more people do not donate, people like Hui will die because of the unavailability of organ transplants.



primarily on statistical information to discuss the implications of organ donation” than were those exposed to the exemplar messages ($M=2.16$, $SD=1.27$; $t_{187}=10.90$, $p<.01$).

RESULTS

Descriptive statistics for the intention scores by gain/loss frame and statistical/exemplar appeal are presented in Table 2. The ANOVA was performed to address Hypothesis 1, and the results are shown in Table 3.

The ANOVA showed no significant main effect of the gain- and loss-framed messages on the intention scores. In contrast, a significant main effect of the statistical/exemplar messages was found ($F_{1,185}=18.20$, $p<.05$; $\eta^2_p=0.09$). Participants exposed to the exemplar message ($M=4.77$, $SD=0.89$) were significantly more likely to express a behavioral intention to become an organ donor than were those exposed to the statistical message ($M=4.15$, $SD=1.19$).

Furthermore, a significant interaction ($F_{1,185}=3.91$, $p=.05$; $\eta^2_p=0.02$) was also found. Specifically, *post hoc* tests of simple main effects showed that the

TABLE 2
DESCRIPTIVE STATISTICS FOR INTENTION SCORES BY GAIN/LOSS
FRAMES AND STATISTICAL/EXEMPLAR APPEALS (N = 189)

Appeal Condition	Frames					
	Gain		Loss		Total	
	M	SD	M	SD	M	SD
Statistical	4.41	1.14	3.84	1.20	4.15	1.19
Exemplar	4.76	0.79	4.79	0.98	4.77	0.89
Total	4.58	1.00	4.33	1.19		

exemplar appeal was more effective, with a moderate effect size, than was the statistical appeal among participants who viewed these under the loss-framed condition ($F_{1,58} = 16.87$, $p < .01$; $\eta^2_p = 0.16$). No differences in participants' intention scores in response to the statistical and exemplar appeals were observed among participants who viewed them under the gain-framed condition. Overall, Hypothesis 1, which predicted a significant interaction effect of message framing and statistical/exemplar appeals on behavioral intentions toward organ donation, was supported by the results of this study.

DISCUSSION

Focusing on the issue of organ donation, this study provided empirical evidence that extends existing research on message framing and exemplar effects. In terms of the gain/loss frame, the results of this study did not support the effect of message framing reported by Reinhart, *et al.* (2007) and Purewal and van den Akker (2010). Three separate experiments were conducted by Reinhart, *et al.* (2007), and the results showed that gain-framed messages were more effective than were loss-framed messages for increasing participants' intentions to donate an organ. Purewal and van den Akker (2010) showed that although the effects of message framing on donating oocytes did not extend to women from South-

TABLE 3
ANALYSIS OF VARIANCE OF SCORES FOR BEHAVIOR INTENTION

Source	Type III SS	df	MS	F	p	η^2_p
Gain/Loss frames (F)	3.50	1	3.50	3.23	.07	0.02
Statistical/Exemplar appeals (A)	19.72	1	19.72	18.20	< .01	0.09
(F) × (A)	4.24	1	4.24	3.91	.05	0.02
Error	200.36	185	1.08			
Total	3,982.13	189				
Corrected total	226.54	188				

east Asia, British female participants under the gain-framed condition were significantly more likely to report higher post-message intentions than were those under the loss-framed condition. Purewal and van den Akker (2010) argued that differences in the ethnicity between Eastern and Western participants may explain these discrepancies. However, previous research regarding Western participants' intentions toward organ donation (Cohen, 2010) also showed that gain frames were not more effective than loss frames in promoting the intention to become an organ donor. Notably, Cohen (2010) pointed out that the messages used in her study focused on vivid exemplars without any explicitly persuasive elements, whereas the Reinhart, *et al.* (2007) study used a direct appeal for organ donation; this difference was cited as the likely the reason for the discrepant results. The messages used in the present study were more similar to the messages used in the study conducted by Reinhart, *et al.* (2007), but the effects of gain/loss frames were not significant. Consequently, the ethnicity factor raised by Purewal and van den Akker (2010) may mediate the effects of message framing on Taiwanese participants' intentions toward organ donation.

In contrast, the results of the present study showed that the statistical/exemplar factor significantly influenced the participants' intentions regarding organ donation. The exemplar messages had a moderate effect size in their influence on Taiwanese college students ($r = .28$). Most studies evaluating the effectiveness of statistical/exemplar appeals on organ donation have found that exemplar messages were more effective than statistical ones (Feeley, *et al.*, 2006; Weber & Martin, 2006; Studts, *et al.*, 2010), and the results of this study were consistent with those previous findings. Kopfman, *et al.* (1998) called attention to the finding that exemplar messages rely on the emotional impact or vividness of the case report to achieve the desired effect. Exemplar messages may be easier to relate to and more memorable, which may influence both subsequent recall of the information they contain and emotional reactions to it. However, emotional arousal in response to exemplar messages was initially strong but declined after a few days. The present study assessed behavioral intentions immediately after the participants read the exemplar messages, and the authors did not measure whether the effects persisted. If not, it is important to identify strategies that extend the effects of exemplar messages long enough to obtain the commitment of recipients to donate organs in the future. For example, frequent emotional messages citing a variety of cases may be effective for this purpose.

This study manipulated gain/loss frames and statistical/exemplar appeals to determine whether four different types of messages would have a different effect on the participants' intentions regarding organ donation. The

results of this study were partially consistent with those of the previous study examining the effect of the same two factors on encouraging Korean participants to donate organs after death (Sun, in press). In the study conducted by Sun (in press), participants who read gain-framed messages with rational appeals showed a more positive attitude than those who read gain-framed messages with emotional appeals. In contrast, participants who read loss-framed messages with emotional appeals showed a more positive attitude than those who read loss-framed messages with rational appeals. The results of the present study also showed that the interaction effect on the participant's intention to donate an organ was statistically significant. However, the interaction effect was observed only in loss-framed message groups. Participants who read loss-framed messages with exemplar appeals were significantly more likely to express positive intentions toward organ donation than were those who read loss-framed messages with statistical appeals, as no significant difference was observed between the statistical and exemplar appeals under the gain-framed condition. Taken together, the descriptive statistics for the intention scores on the four types of messages in the present study show that participants who read loss-statistical messages expressed the least positive intentions toward becoming an organ donor. That is, the persuasiveness of the graphics and the text of the materials used to promote organ donation may not make a significant contribution to the intention to donate an organ when embedded in loss-statistical messages compared with when they are embedded in the other three types of messages.

Due to the enormous advances in surgical techniques and immunology during the last few decades, organ transplantation has become a successful procedure. Nevertheless, the number of transplants is very low in many countries. Unless the number of organ donors increases dramatically, many people will die waiting for a potentially life-saving organ transplant. Increasing the number of viable organs can prevent these deaths. Research has suggested that organ donation behaviors can be changed as a result of various health promotion efforts (Stefanone, *et al.*, 2012; Chien, 2014). Therefore, this study attempted to provide empirical evidence to extend existing research on the effects of message framing and statistical/exemplar appeals to help shape successful organ donation campaigns. In summary, a mixed pattern was observed in this study. Specifically, each of two combinations, one of a gain frame with statistical/exemplar appeals and the second of a loss frame with exemplar appeals, had its unique advantages in predicting behavioral intentions toward organ donation. The results of this study provide a practical guideline for developing an organ donation message for an effective public communication campaign. This study's results can be used as a basis not only for advertisements but also for other forms of promotional material, such as leaflets and pamphlets,

designed to encourage organ donation in Taiwan by using more persuasive and effective messaging.

Limitations and Future Studies

There are several limitations of this study. Because organ donation is a pro-social behavior, the framed messages designed for this study focused on benefits to potential organ recipients. Additional research on the effects of framed messages and exemplar appeals that emphasize recipients' incentives to become organ donors is needed, such as making someone proud, modeling the behavior of an admired other, or feeling good about oneself; this may help researchers extend the effect of message framing on the promotion of organ donation. Moreover, the issue of cultural differences should be examined in future studies because the persuasive effects of message framing and statistical/exemplar appeals may differ across cultures. The final limitation of the present study is that intentions were measured at the time of reading the message, and data on actual behavior were not gathered. Future research should include behavioral measures of responses, such as actually signing donor cards.

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Author Queries

AQ1: Nisbett & Ross, 1980 is not in the references.

AQ2: Kazoleas, 1993 is not in the references.